Model diagnostics

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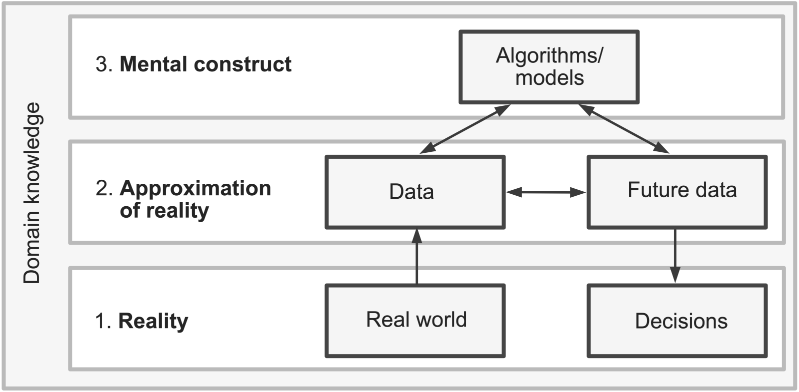
Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Question: Is our model useful?

(Image from *Veridical Data Science*, Yu and Barter)

Motivating example: Dengue data

Data: Data on 5720 Vietnamese children, admitted to the hospital with possible dengue fever. Variables include:

- *Sex*: patient's sex (female or male)
- *Age*: patient's age (in years)
- *WBC*: white blood cell count
- *PLT*: platelet count
- other diagnostic variables...
- *Dengue*: whether the patient has dengue ($0 =$ no, $1 =$ yes)

Motivating example: Dengue data

Y_i = dengue status (0 = negative, 1 = positive)

$$Y_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 WBC_i$$

Assumptions and diagnostics

What assumptions do we make in a *linear* regression model?

Logistic regression assumptions

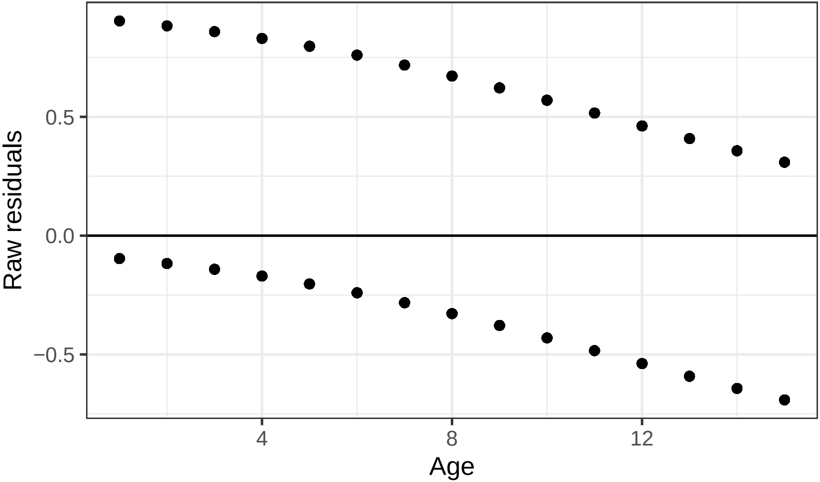
$$Y_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 Age_i$$

Don't use raw residuals for logistic regression

Raw residuals: $Y_i - \widehat{\pi}_i$

Example: predicting dengue status with Age

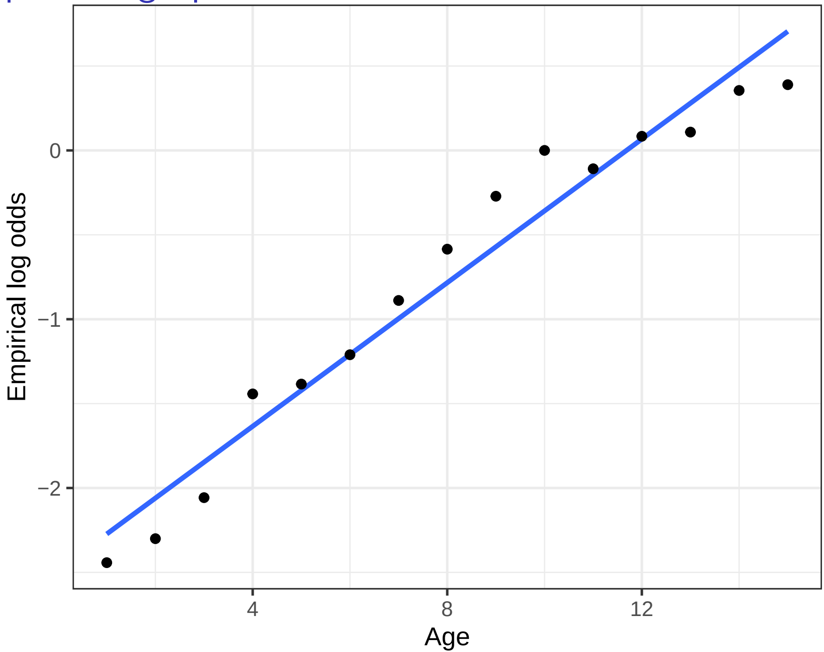


Alternative: empirical logits

For each Age, calculate the *empirical* log odds of dengue (aka *empirical logits*):

```
## # A tibble: 15 x 5
##      Age NoDengue YesDengue      p log_odds
##    <int>    <int>     <int>  <dbl>    <dbl>
##  1     1      184        16 0.08     -2.44
##  2     2      419        42 0.0911   -2.30
##  3     3      532        68 0.113    -2.06
##  4     4      457       108 0.191    -1.44
##  5     5      487       122 0.200    -1.38
##  6     6      369       110 0.230    -1.21
##  7     7      360       148 0.291    -0.889
##  8     8      289       161 0.358    -0.585
##  9     9      244       186 0.433    -0.271
## 10    10      185       185 0.5       0
## 11    11      165       148 0.473    -0.109
## 12    12      126       137 0.521     0.0837
## 13    13       96       107 0.527     0.108
```

Empirical logit plot



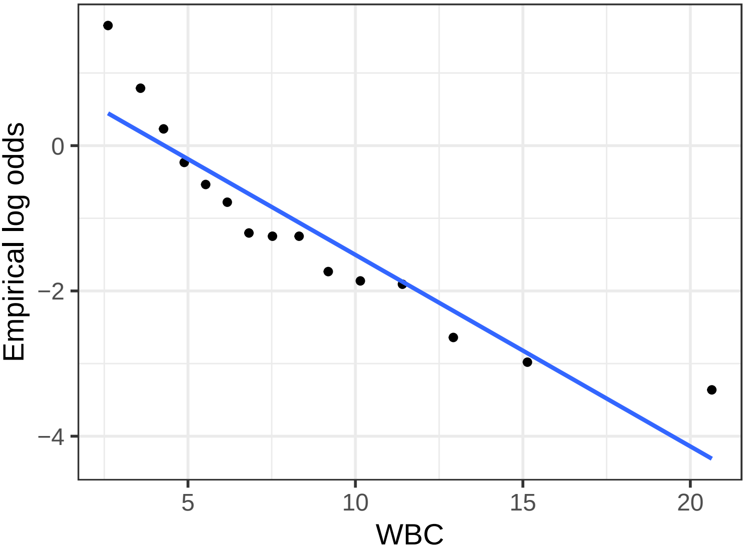
Empirical logits with a continuous variable

Trying with WBC:

```
## # A tibble: 6 x 5
##     WBC NoDengue YesDengue     p log_odds
##   <dbl>    <int>     <int> <dbl>    <dbl>
## 1  0.9         0         1     1      Inf
## 2  1.15        0         2     1      Inf
## 3  1.23        0         1     1      Inf
## 4  1.25        0         1     1      Inf
## 5  1.54        0         3     1      Inf
## 6  1.58        0         1     1      Inf
```

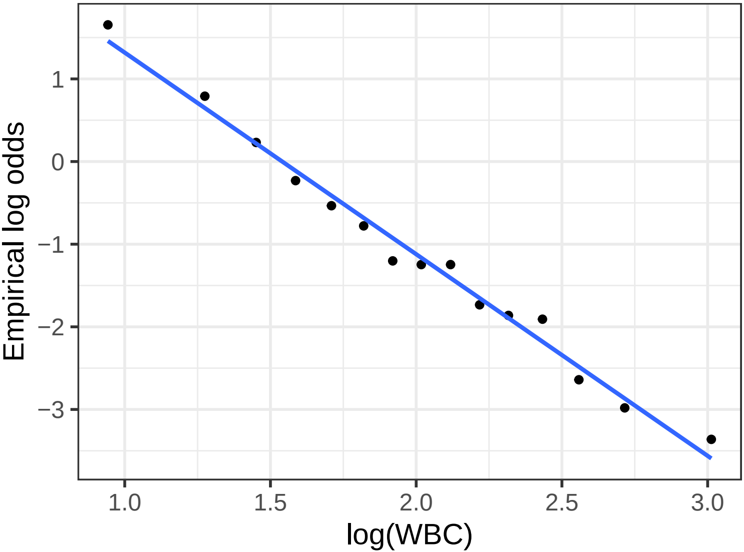
What has gone wrong? What should we do differently?

Binned empirical logit plots

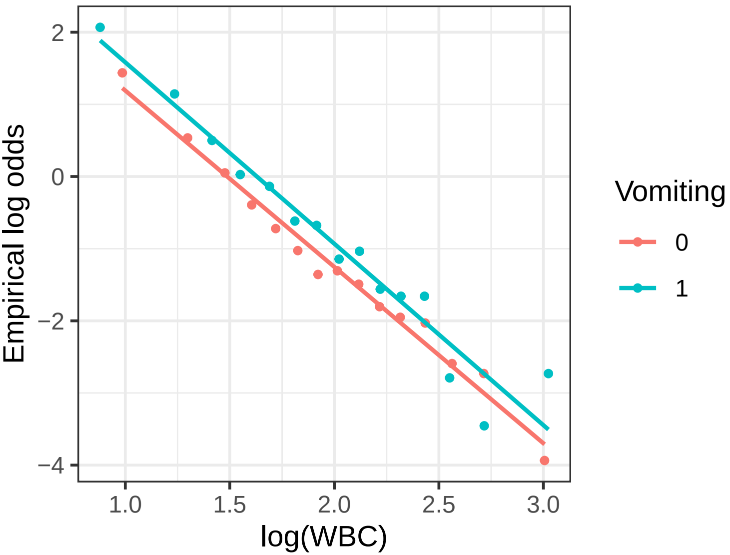


Does the shape assumption seem reasonable?

Binned empirical logit plots



Binned empirical logit plots



Limitations of empirical logit plots

- Does not assess a fitted model (performed *before* fitting the model)
- One point per *bin*, not one point per *observation*
- Can only plot one or two explanatory variables at a time

Class activity